



Serum miRNA biomarker discovery for placenta accreta spectrum

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ABSTRACT

Placenta accreta spectrum (PAS) disorder is a major cause of maternal and fetal morbidity, and *in vitro* biomarkers are highly desired in clinic. This study enrolled three phases of 186 pregnant women, including controls, PAS patients, placenta previa (PP) patients, and pre-eclamptic (PE) patients. Initial miRNA array screened 42 out of 768 serum miRNAs in the screening phase, and then validated four miRNAs by quantitative RT-PCR in the training phase and validation phase. Their performance for PAS prenatal screening was analyzed by the receiver operating characteristic (ROC) curve, sensitivity, and specificity. Data validated that four miRNAs (miR-139-3p, miR-196a-5p, miR-518a-3p, and miR-671-3p) were down-regulated in PAS group comparing with controls in three phases of subjects. Except for miR-518a-3p, the expression levels of these miRNAs also were significantly different between the PAS and the group including PP and PE. In addition, these biomarkers demonstrated modest screening efficiency, as the AUC ranged from 0.59 to 0.74, sensitivity 0.54 to 0.80, and specificity 0.62 to 0.76. However, the AUC and specificity can improve greatly (AUC 0.91, specificity 0.92) using a 'diagnostic signature' that combined the four miRNAs and four clinical parameters into one panel. GO and KEGG signaling pathway analysis indicated their target genes were involved in angiogenesis, embryonic development, cell migration and adhesion, and tumor-related pathways. In conclusion, the four miRNAs discovered in this study not only can be used for future non-invasive prenatal PAS screening, but also provide a new experimental basis for future research on PAS etiology.

1. Introduction

Placenta accreta spectrum (PAS) disorder is a pregnancy complication in which placental villi abnormally adhere to or invade the myometrium [1]. According to statistics, the incidence of PAS in pregnant women is 0.01%–1.1% globally, and the mortality rate is about 0.05%, which seriously endangers the life of the mother and fetus [2]. Studies have shown that placenta previa and cesarean section are major independent factors leading to the onset of PAS, and the increased incidence of these factors is one of the important reasons for the gradual increase of PAS incidence in recent years [3,4]. In mainland China, the caesarean section (CS) rate once reached as high as 62.5% [5], and the family planning initiative launched in 2015 has greatly elevated the number of women in labor with histories of CS, therefore partially contributed to the recent rapid PAS incidence [3].

As PAS is a high-risk condition with serious associated morbidities,

PAS pregnant women should be transferred to a regional medical center immediately for adequate monitoring and timely treatment [2,6]. Because PAS lacks typical clinical manifestations, physical signs and laboratory indicators for effective screening, it is estimated that about 1/2 to 2/3 of the cases remain undiagnosed before delivery [7,8]. The resulting delay in treatment may be one of the key reasons of serious adverse pregnancy outcomes caused by PAS [8].

Generally, high-risk factors of maternal PAS, combined with a prenatal ultrasound or magnetic resonance imaging (MRI), are used for diagnosis [9,10]. However, such imaging diagnosis requires relatively high expense and has technical limitations. Prenatal ultrasound may have false negative report for PAS cases that have low implantation sites and are implanted in the posterior uterus wall [9,10], and contrast agents are rarely used in pregnant women. In addition, the pathological analysis of the separated placenta after delivery, which is the gold standard for PAS diagnosis, cannot provide timely results, and also has the possibility of a

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false negative caused by incomplete separation of the placenta from the uterus [11]. Therefore, there is an urgent need to discover biomarkers for prenatal PAS screening, which are essential for early detection and timely intervention of PAS to reduce mortality.

In this study, we enrolled three groups of individuals, including PAS patients, patients with diseases similar to PAS, and normal pregnancy controls in a provincial centre for PAS women. The differentially expressed serum miRNA molecules were detected and further validated in these three groups, and the corresponding target genes and signal pathways were further analyzed for PAS biomarker discovery.

2. Materials and methods

2.1. Study design

This study was approved by the Ethics Committee of the Third Affiliated Hospital of Guangzhou Medical University (No.2014[085]) and written consent was obtained from all participants. Cases and controls were pregnant women who delivered at the Provincial Centre for Critical Pregnant Women at the Third Affiliated Hospital of Guangzhou Medical University from August 2015 to December 2018. All PAS cases were diagnosed according to the diagnostic criteria using intraoperative diagnosis and postpartum histopathological analysis [12], and only cases of placenta increta and placenta percreta during their third trimester of pregnancy were included. Controls (CON) were healthy pregnant women during their third trimester of pregnancy with a single fetus and without any pregnant complications. Simultaneously, we enrolled pre-eclamptic (PE) and placenta previa (PP) patients during their third trimester of pregnancy with single fetuses. Clinical parameters, such as maternal age; BMI; number of previous gravidity, parity, and CS; gestational week at birth; diastolic and systolic blood pressure (DBP, SBP); incidence of proteinuria; blood loss during delivery; birth weight, and Apgar score at 10 min, were clinically recorded. In total, this study enrolled three groups of 186 pregnant women, including 12 PAS patients and 12 CON women in the screening phase, 41 PAS patients and 41 CON women in the training phase, and 20 PAS patients, 20 PP patients, 20 PE patients and 20 CON women in the validation phase (Supplemental Fig. 1).

2.2. Serum preparation

Venous blood of all subjects was collected using a non-anticoagulant tube during the third trimester before delivery of all pregnant women. The blood sample was placed at room temperature for 30 min, centrifuged at 2000×g for 15 min to separate serum and blood cells and stored separately in −80 °C freezer in the Biobank of the Third Affiliated Hospital of Guangzhou Medical University until subsequent studies.

2.3. RNA isolation

Total RNA was extracted from 200 µl of serum using the MagMAX™ mirVana™ Total RNA Isolation Kit (ABI, Carlsbad, California, USA) and was eluted with 50 µl of RNase-free H₂O. The extracted RNA was stored at −80 °C until subsequent studies.

2.4. Microarray screening

In the screening phase, 2.5–5.0 ng of RNA from each sample was reverse transcribed using the miRCURY Locked Nucleic Acid (LNA™) Universal Reverse Transcription (RT) microRNA PCR, Polyadenylation and cDNA synthesis kit (Exiqon miRNA qPCR panel, Vedbaek, Denmark), according to the manufacturer's protocol. A total of 768 miRNAs in the sample were detected by the 7900HT real-time PCR system (Applied Biosystems, Foster City, CA, USA) with Exiqon miRCURY-Ready-to-Use PCR-Human-panel-I + II-V4.M (Exiqon miRNA qPCR panel, Vedbaek, Denmark). Among them, miRNAs with a cycle

threshold (Ct) of less than 37 and 5 Ct values below the negative control were analyzed.

2.5. Quantitative RT-PCR (qRT-PCR)

The target miRNA and reference miRNA (U6, hsa-miR191-5p and hsa-miR-423-5p) of each sample was specifically reverse transcribed using the corresponding primer (Supplemental Table 1) with Prime-Script RT Master Mix (Perfect Real Time, Takara, Japan) according to the manufacturer's instructions. The level of miRNA was detected using 1 µl cDNA by qRT-PCR with TB Green™ Premix Ex Taq™ II (TaKaRa, Japan) and expression level of target miRNA was normalized using reference miRNA.

2.6. Statistical analysis

All data was analyzed using SPSS 20.0 software (SPSS IBM, New York, USA) and GraphPad Prism 7.0 software (GraphPad Software, San Diego, CA, USA). Comparisons between groups were performed using the Mann–Whitney test in the nonparametric data, and the unpaired *t*-test was used to analyze numeric data. The diagnostic value of miRNA was analyzed using the receiver operating characteristic (ROC) curve, sensitivity, specificity, positive prediction value, negative prediction value, and Youden index, and *P* < 0.05 was considered as statistically significant.

2.7. Bioinformatics analysis

Target gene prediction was performed using two databases, DIANA and Targetscan. The overlapped prediction portion from the two databases served as the target genes for final prediction. The obtained target genes were performed using Gene ontology (GO) and signal pathway analysis, of which, the GO analysis included three categories: biological process (BP), cellular component (CC) and molecular function (MF). Signal pathway analysis was performed by matching the predicted target genes with the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways for functional analysis. The obtained results were subjected to the Fisher's exact test, and a lower *p* value represented a higher importance of the GO term or pathway.

3. Results

3.1. Clinical findings

The clinical characteristics of the case group patients and CON women are listed in Table 1. Among them, the number of previous gravidity, parity, and CS in the PAS group was significantly higher than those in the CON group (*P* < 0.05, Table 1). In addition, the PAS group had significantly increased blood loss during delivery and hysterectomy percentage (*P* < 0.001, Table 1). The admission blood pressure and urine protein positive rate in the PE group were significantly higher than those in the CON group, the PAS group and the PP group (*P* < 0.05, Table 1). The gestational age and body weight of the newborns in the PAS, PE and PP groups were significantly lower than those in the CON group (*P* < 0.05, Table 1).

3.2. Identification and validation of disease-associated miRNAs

In the screening phase, 42 miRNAs with a *p* value of less than 0.05 or with a fold change of more than 2 were screened from 768 miRNAs using microarray (Supplemental Table 2). In the training phase, five miRNAs (miR-122-5p, miR-139-3p, miR-196a-5p, miR-518a-3p, miR-671-3p) with a *p* value of less than 0.05 were identified from the 42 miRNAs screened in the screening phase using qRT-PCR (Table 2).

In the validation phase, the expression level of miRNA-139-3p, miR-196a-5p, and miRNA-671-3p in PAS group significantly decreased

Table 1
Clinical characteristics of the subjects in the three phases.

Characteristics	Screening phase		Training phase		Validation phase			
	CON	PAS	CON	PAS	CON	PAS	PP	PE
Age (yrs)	27.75 ± 5.15	29.00 ± 3.41	30.32 ± 4.80	33.93 ± 3.80	29.95 ± 4.39	31.85 ± 5.42	33.05 ± 4.10	32.10 ± 3.74
BMI (kg/m ²)	21.60 ± 2.40	21.46 ± 3.83	20.93 ± 3.01	21.68 ± 2.63	20.08 ± 2.69	20.28 ± 3.04	19.86 ± 1.51	22.30 ± 3.54
Gravidity (n) ^a	1.75 ± 0.60	3.17 ± 1.11*	2.00 ± 1.16	3.41 ± 1.09*	2.55 ± 1.28	3.40 ± 1.54 ^c	2.75 ± 1.33	2.40 ± 1.10
Parity (n) ^a	0.50 ± 0.50	1.33 ± 0.65*	0.49 ± 0.55	1.20 ± 0.62*	0.70 ± 0.80	1.15 ± 0.59*	1.05 ± 0.76	0.70 ± 0.66
Caesarean section (n) ^a	0.33 ± 0.47	1.12 ± 0.62*	0.24 ± 0.43	1.10 ± 0.62*	0.20 ± 0.41	1.05 ± 0.69*	0.75 ± 0.44*	0.50 ± 0.61
Systolic blood press (mmHg) ^b	113 ± 9	109 ± 8	116 ± 10	114 ± 9	113 ± 8	108 ± 11	114 ± 11	162 ± 36*
Diastolic blood press (mmHg) ^b	69 ± 8	68 ± 9	74 ± 7	70 ± 8	72 ± 7	71 ± 7	71 ± 6	103 ± 29*
Urine Protein (g/24 h) ^b	0	0	0	0.11 ± 0.08	0	0	0	2.12 ± 2.69*
Intraoperative report (%) ^c	0	100	0	100	0	100	0	0
Pathological diagnosis (%) ^c	0	100	0	100	0	100	0	0
Bleeding during delivery (ml)	207 ± 55	1267 ± 781**	242 ± 80	2083 ± 1937**	296 ± 121	1550 ± 1262**	440 ± 490	329 ± 149
Hysterectomy (%) ^d	0	8.33	0	24.39	0	25.00	0	0
Gestational week at birth (wks) ^b	39.94 ± 0.85	36.49 ± 1.64*	39.68 ± 0.84	35.21 ± 2.31*	39.88 ± 0.78	35.39 ± 2.98*	35.85 ± 2.34*	35.42 ± 3.18*
Birth weight (g) ^b	3390 ± 343	2848 ± 577*	3241 ± 363	2560 ± 534*	3397 ± 508	2479 ± 610*	2565 ± 499*	2100 ± 997*
Apgar(10min) ^b	10.00 ± 0	10.00 ± 0	10.00 ± 0	9.95 ± 0.22	10.00 ± 0	9.95 ± 0.22	9.95 ± 0.22	9.75 ± 0.64

Note: Data are present as Mean ± S.D.

CON, controls. PAS, placenta accreta spectrum disorders. PP, placenta previa. PE, preeclampsia.

^a The number of previous gravidity, parity, and caesarean section. Data were compared by M-W test.

^b Data were compared by unpaired *t*-test.

^c The percentage of cases diagnosed as PAS by surgeon or pathological analysis.

^d The percentage of cases that had hysterectomy.

^e *P* < 0.1, **P* < 0.05, and ***P* < 0.001 compared to CON group.

Table 2
Expression changes of miRNAs in the training phase.

miRNA ID	PAS/CON	<i>p</i> value
miR-122-5p	3.6263	0.02 ^a
miR-139-3p	0.6625	0.02 ^a
miR-196a-5p	0.5162	0.04 ^a
miR-518a-3p	0.4983	0.02 ^a
miR-671-3p	0.5328	0.01 ^a

Note: Data are present as the fold of PAS to CON groups. CON, controls. PAS, placenta accreta spectrum disorders.

^a *P* < 0.05 compared to CON group by unpaired *t*-test.

compared with other groups (*P* < 0.05, Table 3). Although the expression of miRNA-518a-3p was not significantly different between the PAS group and the PE groups, it decreased compared with that of the CON group and PP group (*P* < 0.05 or *P* < 0.1, Table 3).

To evaluate the value of the four target miRNAs in the diagnosis of PAS, all sample data were regrouped, and the PP and PE groups were combined as the NON-PAS group. Compared with the CON group and NON-PAS group, the expression level of miRNA-139-3p, miR-196a-5p and miRNA-671-3p in PAS patients decreased (*P* < 0.05, Fig. 1A and B, D). The expression of miRNA-518a-3p also decreased in the PAS group compared with the CON group (*P* < 0.05, Fig. 1, C), while it was not significantly different between the PAS and NON-PAS group.

Table 3
Expression changes of miRNAs in different groups of the validation phase.

miRNA ID	PAS VS. CON		PAS VS. PE		PAS VS. PP	
	Ratio	<i>p</i> value	Ratio	<i>p</i> value	Ratio	<i>p</i> value
miR-139-3p	0.63	0.01 ^b	0.64	0.02 ^b	0.72	0.05 ^b
miR-196a-5p	0.55	0.01 ^b	0.57	0.01 ^b	0.57	0.01 ^b
miR-518a-3p	0.51	0.04 ^b	0.76	0.21	0.57	0.06 ^a
miR-671-3p	0.65	0.04 ^b	0.66	0.06 ^a	0.59	0.03 ^b

Note: Data are present as the fold of PAS to CON, P or PE groups. CON, controls. PAS, placenta accreta spectrum disorders. PP, placenta previa. PE, preeclampsia.

^a *P* < 0.1 by unpaired *t*-test.

^b *P* < 0.05 by unpaired *t*-test.

3.3. Value of miRNAs for PAS screening

Using bivariate correlation analysis, it was found that the incidence of PAS was negatively correlated with the expression of miR-139-3p, miR-196a-5p, and miR-671-3p, with correlation coefficient of −0.414, −0.420, −0.352, respectively (all *P* < 0.0001). The amount of blood loss was negatively correlated with the expression of miR-139-3p, and miR-196a-5p, with correlation coefficient of −0.215 and −0.172, respectively (*P* < 0.01 and *P* < 0.05). Hysterectomy was positively correlated with the amount of blood loss, and correlation coefficient was 0.426 (*P* < 0.0001).

Using logistic model, the performance of the four miRNAs individually and combined miRNAs in PAS screening was analyzed in all populations, and the area under the curve (AUC), sensitivity, specificity, positive prediction value, negative prediction value and the Youden index were calculated (Table 4, Fig. 1). In the combination model, [miR-139], [miR-196a-5p], [miR-518a-3p] and [miR-671] were used to represent their serum expression levels, and [gravidity], [parity], [CS], and [gestational week at sampling] were the value of previous gravidity, parity, CS, and gestational week. This 'diagnostic signature' was calculated using the logistic regression method as following: EXP(6.302 - 0.834*[miR-139] - 0.44*[miR-196a-5p] + 0.427*[miR-518a-3p] + 0.415*[miR-671] + 0.604*[gravidity] - 0.843*[parity] + 2.733*[CS] - 0.223*[gestational week at sampling]) / {1 + EXP(6.302 - 0.834*[miR-139] - 0.44*[miR-196a-5p] + 0.427*[miR-518a-3p] + 0.415*[miR-671] + 0.604*[gravidity] - 0.843*[parity] + 2.733*[CS] - 0.223*[gestational week at sampling])}.

The results showed that miR-139-3p had the highest sensitivity (0.80). However, the combined miRNAs and clinical parameters had the highest AUC value (0.91), specificity (0.92), positive prediction value (0.88), negative prediction value (0.85) and Youden index (0.69). Therefore, when the four miRNAs and four clinical parameters were combined into a panel, a significantly better diagnostic performance was obtained compared with usage of single miRNA (Table 4, Fig. 1).

3.4. Bioinformatics analysis of target miRNA

A total of 682 target genes were predicted using DIANA and Targetscan databases, including 93 target genes of miR-139-3p, 506 of miR-196a-5p, 22 of miR-518a-3p and 61 of miR-671-3p. GO analysis of these

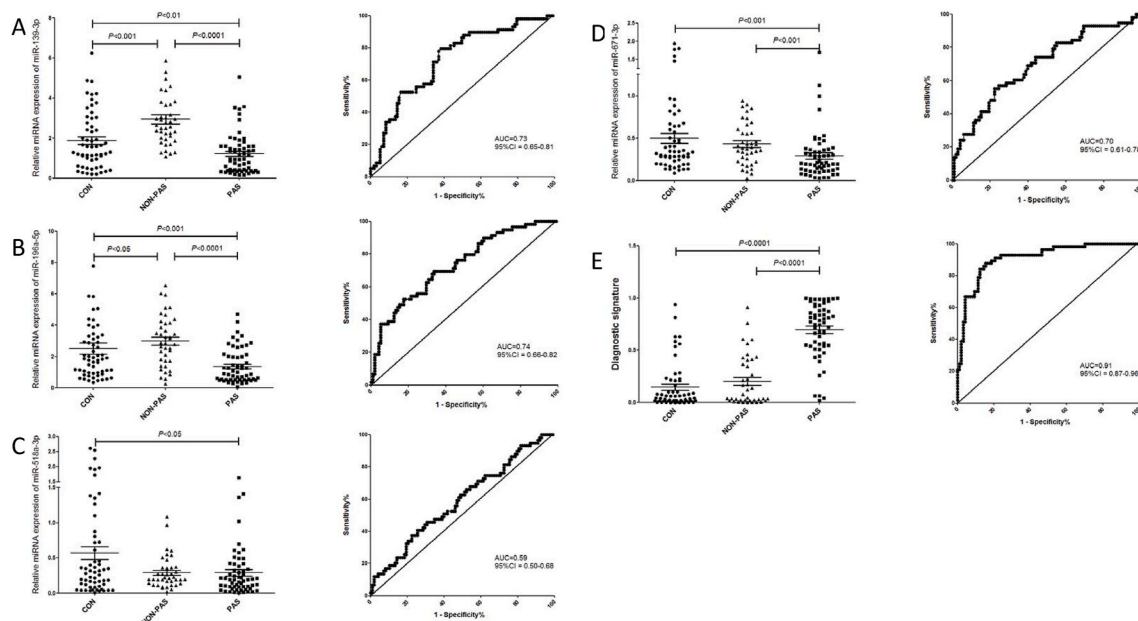


Fig. 1. Expression levels (left) and ROC curves (right) of miRNAs in all subjects. (A) miR-139-3p; (B) miR-196a-5p; (C) miR-518a-3p; (D) miR-671-3p; (E) diagnostic signature. CON, controls. PAS, placenta accreta spectrum disorders. NON-PAS, combination of placenta previa and preeclampsia. ROC, receiver operating characteristics.

Table 4
Diagnostic performance of miRNAs in all subjects.

miRNA ID	AUC(95%CI)	Sensitivity	Specificity	PPV	NPV	Youden index
miR-139-3p	0.73 (0.65–0.81)	0.80	0.62	0.56	0.83	0.41
miR-196a-5p	0.74(0.66–0.82)	0.69	0.66	0.55	0.78	0.36
miR-518a-3p	0.59 (0.50–0.68)	0.54	0.69	0.52	0.72	0.24
miR-671-3p	0.70(0.61–0.78)	0.57	0.76	0.58	0.75	0.32
Combination ^a	0.91 (0.87–0.96)	0.77	0.92	0.88	0.85	0.69

AUC: area under curve; PPV: positive prediction value; NPV: negative prediction value.

^a Calculation method was described in the Results section.

genes showed enrichment in MF included nucleotide binding, protein binding, phosphodiesterase activity and MAP kinase activity. Genes enriched in the CC include membrane, cytoplasm, nucleoplasm and receptor complex proteins. Genes enriched in BP include embryonic skeletal system formation, protein expression regulation, and cell differentiation, movement and migration factors (Fig. 2, A). KEGG pathway analysis revealed that the predicted target genes are involved in cancer, focal adhesion, glioma and the ErbB signaling pathway (Fig. 2, B).

4. Discussion

In this study, we conducted a case-control study in a regional maternal critical care center for PAS patients, and three phases of cases and controls were recruited and differentially expressed serum miRNA molecules were analyzed and validated. The data showed that the expression of miR-196a-5p, miR-518a-3p, miR-139-3p and miR-671-3p in PAS patients was significantly downregulated. In addition, they demonstrate a significant discrimination degree in abnormal placentation associated diseases including PAS, PE and PP. Further bioinformatics revealed that these four miRNAs are related to cell migration and adhesion, and tumorigenesis. As the prevalence and incidence of PAS are rapidly becoming a critical issue [13], the obtained miRNA biomarkers in this study, with high operability, acceptable sensitivity and specificity, and low cost, might be put into clinical application soon [14].

Our data identified that the expression of miR-196a-5p, miR-518a-3p, miR-139-3p and miR-671-3p in PAS patients was significantly downregulated. Further analysis demonstrated that single miRNA could

be used to screen PAS with modest efficiency, as indicated by the AUC ranged from 0.59 to 0.74, sensitivity ranged from 0.54 to 0.80, and specificity ranged from 0.62 to 0.76. However, the AUC and specificity improved greatly using the 'diagnostic signature' that combined four miRNAs and four clinical parameters into one panel, and therefore greatly increased the diagnostic performance for clinical practice. In addition, considering that in each sample, the detection of any one of the miRNAs alone may result in a missing value, screening with the combined miRNAs is more stable and reliable for clinic application.

Notably, miR-139-3p, miR-196a-5p, miR-518a-3p and miR-671-3p demonstrated a significant discrimination degree in similar pregnancy diseases (PAS, PE and PP). Among them, expression of miR-139-3p and miR-196a-5p significantly decreased in PAS but increased in NON-PAS group including PE and PP; miR-518a-3p expression decreased in both PAS and NON-PAS groups; miR-671-3p expression decreased only in PAS. These results suggested that these four miRNAs might play different regulatory roles in placental development, therefore they can be used for differential diagnosis of PAS from PE and PP.

Through further GO and KEGG signaling pathway analysis of the target gene functions of these four miRNAs, we found that the functions of a large number of target genes were duplicated, and that most of them were involved in angiogenesis, embryonic development, cell migration and adhesion, and tumor-related pathways, which is consistent with the phenomenon of abnormal adhesion of placental villi and excessive invasion of extravillous trophoblast cells in PAS cases in clinic.

Previous studies have shown that the miRNA family is an important component of the gene expression regulatory network and is involved in

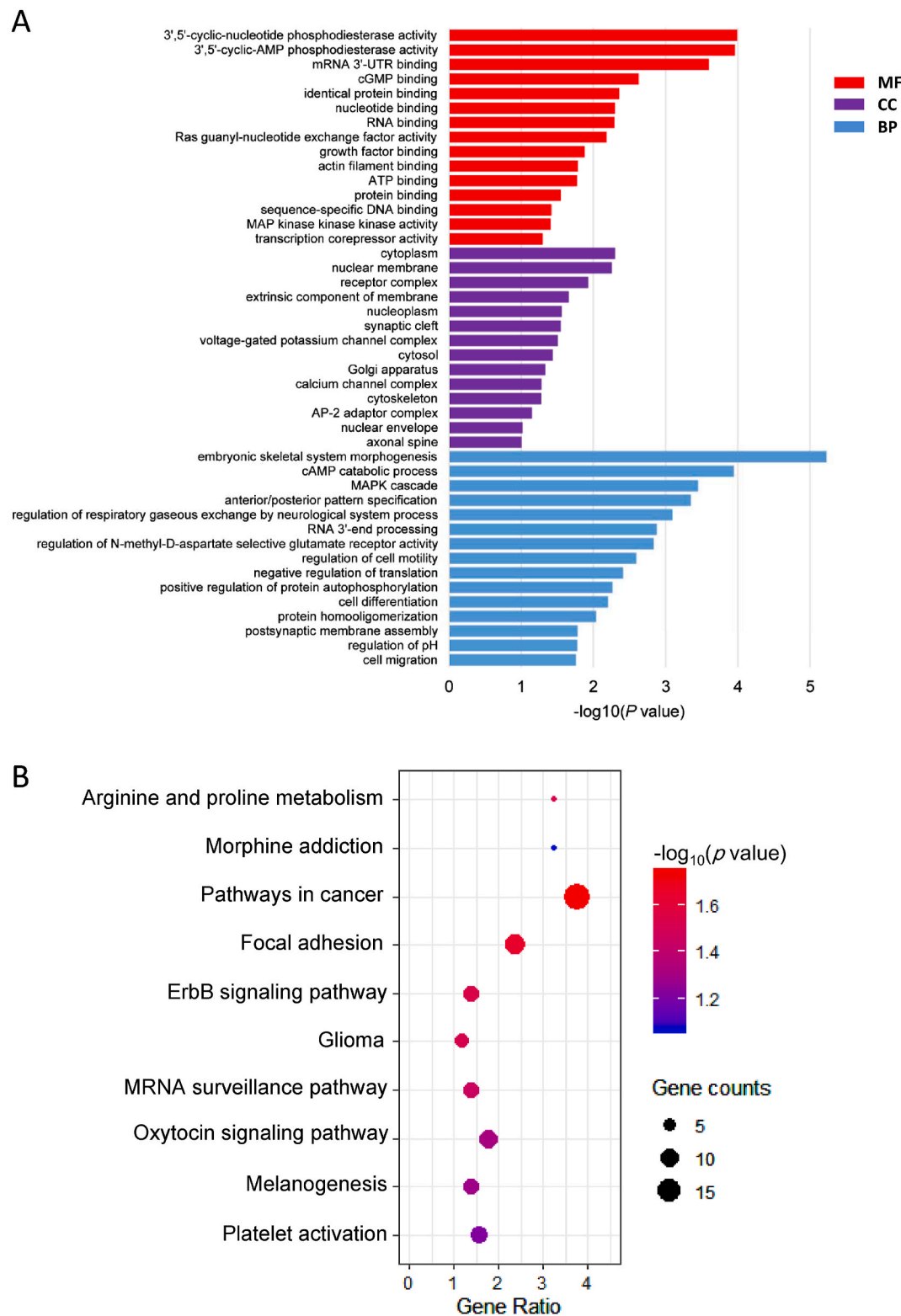


Fig. 2. Bioinformatic analysis of predictive target genes. (A) Gene ontology (GO) analysis of target genes (B) analysis of Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway. MF, Molecular Function. CC, Cellular Component. BP, Biological Process.

the regulation of many important signaling pathways [15], and recent researches have reported that circulating miRNAs in the maternal circulation might serve as biomarkers for monitoring pregnancy complications such as preeclampsia [16–19]. Although this study did not further explore the specific mechanisms of these miRNAs related to the occurrence and development of PAS, related reports and biological

analyses both suggest that these four miRNAs are associated with abnormal adhesion of placental villi and excessive invasion of extravillous trophoblast cells in PAS, and also indicate that PAS has certain characteristics of tumorigenesis.

Among these four miRNAs discovered in this study, miR-518a-3p is located in the miRNA (C19MC) cluster on chromosome 19 [19], which is

a primate-specific miRNA profile expressed in the placenta [20]. Functional studies suggest that downregulation of miR-518a-3p directly activates the NF- κ B pathway, an important tumor-related pathway [21, 22]; In addition, downregulated miR-139-3p can promote epithelial-mesenchymal-transition (EMT) by promoting the expression of matrix metalloproteinase 11 [23,24], and can also promote cell proliferation, migration and invasion through multiple pathways [23,25, 26]. Meanwhile, downregulation of miRNA-196a-5p can inhibit cell proliferation, migration and invasion by promoting the expression of the ANXA1 gene [27], and can promote EMT by promoting the expression of E-cadherin [24], which is considered an important factor in the occurrence of PAS [28]. In addition, miR-671-3p has been shown to exhibit low expression levels in some tumors [29].

In clinic, translational research on PAS biomarker is highly limited, as PAS incidence is relatively low and the heterogeneity and uncertainty of PAS diagnosis was frequent [2,30]. Compared with previous studies [31,32], our study was a large case-control study in a provincial center for PAS patients. All cases were diagnosed based on intraoperative reports and postpartum histopathological analysis. Therefore, this study provided potential biomarkers to develop prenatal screening tool for PAS. However, since this is a study in a single center in southern China, confounding and geographic biases are unavoidable. A multiple-centered prospective study is required to further confirm the clinical utility in large populations.

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Declaration of competing interest

There are no competing interests for any author.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.placenta.2020.09.068>.

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